Lord of the Wings [2026]

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Quest 4: The Rankening

‹ Quest 3 Quest 5 ›

Part I

Story section

After what felt like forever, the votes had finally been counted and the new Grand Council was chosen. We explained why we had travelled to the Elven Capital.

Glitchron, a powerful demon from the Undefined Realm, had rebuilt the fortress of Nulldor once again. Long ago, he had cursed the eggs of the ancient Dragonducks, twisting the creatures inside them into his Riders of Darkness. Only one egg had survived to this day: the stone that Frodoff had been carrying since we left Quackshire. Inside was the last descendant of those ancient Dragonducks, but Glitchron's curse was slowly changing what it would become. If the egg ever hatched under his influence, the creature inside would emerge as the Prince of Darkness. The egg could only be destroyed in Nulldor, of course. Frodoff finally understood where his clever idea in Gimlock's forge had come from. The cursed egg was slowly taking control of his mind, just as the curse was taking control of the hatchling inside.

The Grand Council agreed to help our expedition. They wanted to send their best warrior with us, but there was one problem. Nobody knew who the best warrior was. The annual Tournament of Elven Warriors had just finished. The warriors had completed many quests, but nobody had calculated the final standings yet. The winner still had to be determined.

Gimlock crossed his arms and said that if the records contained another ISO standard, he was going back to his mountain. Luckily, this time the records looked much more ordinary.

We were given the results of one of the tournament quests as a trial (your notes). The quest consisted of three parts. Every line described the result of one participant in one part using the following format: `id=7 quest=1 part=2 time=345`

- `id` was the participant's identifier,
- `quest` identified the tournament quest,
- `part` was the part number (1, 2, or 3),
- `time` was the participant's completion time for that part, in seconds.

The list contained the complete results for a single quest. **Every participant had finished all three parts**, so the time for every part of every participant was present exactly once. The records were in no particular order.

We had to **calculate the total time for every participant** by adding together the times from all three parts. The participants then had to be ranked from the shortest total time to the longest.

Finally, we calculated a checksum so that the elves could easily verify our results. Starting from the top of the ranking, every participant's **total time was multiplied by their finishing position** (1 for first place, 2 for second, and so on). The checksum was the **sum of all these products.** Fortunately, nobody finished with the same total time, so the final ranking contained no ties.

Example based on the following notes:

```
id=1 quest=1 part=1 time=187
id=3 quest=1 part=1 time=413
id=5 quest=1 part=3 time=744
id=2 quest=1 part=3 time=745
id=4 quest=1 part=2 time=558
id=1 quest=1 part=2 time=299
id=1 quest=1 part=3 time=362
id=4 quest=1 part=3 time=1268
id=4 quest=1 part=1 time=457
id=5 quest=1 part=1 time=257
id=2 quest=1 part=1 time=192
id=3 quest=1 part=3 time=1168
id=5 quest=1 part=2 time=346
id=3 quest=1 part=2 time=455
id=2 quest=1 part=2 time=562
```

First, we calculated the total time for every participant:

```
id=1: 187 + 299 + 362 = 848
id=2: 192 + 562 + 745 = 1499
id=3: 413 + 455 + 1168 = 2036
id=4: 457 + 558 + 1268 = 2283
id=5: 257 + 346 + 744 = 1347
```

Sorting the participants by their total times gave the following ranking:

```
1st: id=1 (848)
2nd: id=5 (1347)
3rd: id=2 (1499)
4th: id=3 (2036)
5th: id=4 (2283)
```

The checksum was therefore: 1*848 + 2*1347 + 3*1499 + 4*2036 + 5*2283 = `27598` .

What was the checksum of the final ranking?

Your notes for this part:
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Part I solved with answer: 201498
Check your progress

Part II

The elves wanted to be certain that we could calculate the rankings correctly. They gave us the complete results from the previous tournament: all 20 quests and every recorded result for every participant (your notes).

Most participants had not completed every part of every quest. The number of completed parts was the most important value when determining the ranking. **A participant who had completed more parts always ranked higher than one who had completed fewer.**

If two participants had completed the same number of parts, the one with **the shorter total time ranked higher.** We then calculated the checksum exactly as before. The final ranking contained no ties, so every finishing position was unique.

Example based on the following notes:

```
id=2 quest=1 part=1 time=100
id=4 quest=2 part=1 time=150
id=1 quest=2 part=1 time=140
id=1 quest=1 part=2 time=180
id=2 quest=2 part=1 time=130
id=4 quest=1 part=1 time=110
id=1 quest=2 part=2 time=210
id=2 quest=2 part=2 time=240
id=1 quest=2 part=3 time=330
id=2 quest=1 part=2 time=170
id=3 quest=1 part=1 time=90
id=4 quest=1 part=2 time=175
id=3 quest=2 part=1 time=120
id=1 quest=1 part=1 time=120
id=3 quest=2 part=2 time=220
id=2 quest=2 part=3 time=350
id=3 quest=1 part=2 time=160
```

First, we counted the number of completed parts and the total time for every participant:

```
id=1: 5 parts, total time = 980
id=2: 5 parts, total time = 990
id=3: 4 parts, total time = 590
id=4: 3 parts, total time = 435
```

Participants with more completed parts always ranked higher. If two participants had completed the same number of parts, the shorter total time decided the order.

The final ranking was therefore:

```
1st: id=1 (5 parts, 980)
2nd: id=2 (5 parts, 990)
3rd: id=3 (4 parts, 590)
4th: id=4 (3 parts, 435)
```

The checksum was: 1*980 + 2*990 + 3*590 + 4*435 = `6470`

What was the checksum of the final ranking?

Your notes for this part:
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Part II solved with answer: 575619232
Check your progress

Part III

The elves were finally convinced that we could calculate rankings correctly. They handed us the results of the latest tournament (your notes) and asked us to determine the final standings.

This time, the ranking was no longer based directly on the number of completed parts or the total time. **Instead, every completed part awarded ranking points.**

For every quest, a new ranking was calculated after each part. The fastest participant received:

- **50** points for part 1,
- **100** points for part 2,
- **150** points for part 3.

Every following participant received one point fewer, down to 1 point. Any remaining participants received 0 points. Participants who had not completed a part also received 0 points for that part.

One quest was played each day, and the times from its parts were added together when determining these rankings. The ranking after part 2 was **based on the combined time** of part 1 + part 2, while the ranking after part 3 was based on the combined time of part 1 + part 2 + part 3.

If two or more participants had exactly **the same combined time after a given part**, they shared the same position and each received the points for that position. For example, if two participants tied for first place in Part 2, they both received 100 points, while the next participant received 98 points.

After adding together all the points from every completed part, the participants were ranked by their total score. We then calculated the checksum as before, using each participant's final ranking position and total number of points. **If two or more participants finished with the same total score, they shared the same position when calculating the checksum.**

Example based on the following notes:

```
id=2 quest=1 part=1 time=100
id=4 quest=2 part=1 time=150
id=1 quest=2 part=1 time=140
id=1 quest=1 part=2 time=180
id=2 quest=2 part=1 time=130
id=4 quest=1 part=1 time=110
id=1 quest=2 part=2 time=210
id=2 quest=2 part=2 time=240
id=1 quest=2 part=3 time=330
id=2 quest=1 part=2 time=170
id=3 quest=1 part=1 time=90
id=4 quest=1 part=2 time=175
id=3 quest=2 part=1 time=120
id=1 quest=1 part=1 time=120
id=3 quest=2 part=2 time=220
id=2 quest=2 part=3 time=350
id=3 quest=1 part=2 time=160
```

We worked through the results one quest at a time and one part at a time.

For quest 1, the ranking after part 1 was:

```
1st: id=3, time 90  : 50 points
2nd: id=2, time 100 : 49 points
3rd: id=4, time 110 : 48 points
4th: id=1, time 120 : 47 points
```

After part 2, the times from both completed parts were added together:

```
1st: id=3, 90 + 160 = 250  : 100 points
2nd: id=2, 100 + 170 = 270 : 99 points
3rd: id=4, 110 + 175 = 285 : 98 points
4th: id=1, 120 + 180 = 300 : 97 points
```

Nobody completed part 3 of this quest, so no points were awarded for it.

For quest 2, the participants received:

```
Part 1:
1st: id=3, time 120 : 50 points
2nd: id=2, time 130 : 49 points
3rd: id=1, time 140 : 48 points
4th: id=4, time 150 : 47 points

Part 2:
1st: id=3, 120 + 220 = 340 : 100 points
2nd: id=1, 140 + 210 = 350 : 99 points
3rd: id=2, 130 + 240 = 370 : 98 points

Part 3:
1st: id=1, 140 + 210 + 330 = 680 : 150 points
2nd: id=2, 130 + 240 + 350 = 720 : 149 points
```

Adding all the points gave:

```
id=1: 47 + 97 + 48 + 99 + 150 = 441
id=2: 49 + 99 + 49 + 98 + 149 = 444
id=3: 50 + 100 + 50 + 100 = 300
id=4: 48 + 98 + 47 = 193
```

The final ranking was therefore:

```
1st: id=2, 444 points
2nd: id=1, 441 points
3rd: id=3, 300 points
4th: id=4, 193 points
```

The checksum was: 1*444 + 2*441 + 3*300 + 4*193 = `2998`

```
id=2 quest=1 part=1 time=90
id=4 quest=2 part=1 time=150
id=1 quest=2 part=1 time=130
id=1 quest=1 part=2 time=180
id=2 quest=2 part=1 time=130
id=4 quest=1 part=1 time=110
id=1 quest=2 part=2 time=210
id=2 quest=2 part=2 time=220
id=1 quest=2 part=3 time=330
id=2 quest=1 part=2 time=170
id=3 quest=1 part=1 time=90
id=4 quest=1 part=2 time=170
id=3 quest=2 part=1 time=120
id=1 quest=1 part=1 time=120
id=3 quest=2 part=2 time=220
id=2 quest=2 part=3 time=350
id=3 quest=1 part=2 time=170
```

Frodoff insisted that we should also practise handling ties before starting the real task.

For quest 1, part 1, two participants shared first place:

```
1st: id=2, time 90  : 50 points
1st: id=3, time 90  : 50 points
3rd: id=4, time 110 : 48 points
4th: id=1, time 120 : 47 points
```

For quest 1, part 2, the ranking was based on the combined times from parts 1 and 2:

```
1st: id=2, 90 + 170 = 260  : 100 points
1st: id=3, 90 + 170 = 260  : 100 points
3rd: id=4, 110 + 170 = 280 : 98 points
4th: id=1, 120 + 180 = 300 : 97 points
```

Nobody completed part 3 of this quest, so no points were awarded for it.

For quest 2, part 1, two participants shared second place:

```
1st: id=3, time 120 : 50 points
2nd: id=1, time 130 : 49 points
2nd: id=2, time 130 : 49 points
4th: id=4, time 150 : 47 points
```

For quest 2, part 2, two participants shared first place:

```
1st: id=1, 130 + 210 = 340 : 100 points
1st: id=3, 120 + 220 = 340 : 100 points
3rd: id=2, 130 + 220 = 350 : 98 points
```

Only two participants completed quest 2, part 3:

```
1st: id=1, 130 + 210 + 330 = 670 : 150 points
2nd: id=2, 130 + 220 + 350 = 700 : 149 points
```

Adding together all the points gave:

```
id=1: 47 + 97 + 49 + 100 + 150  = 443
id=2: 50 + 100 + 49 + 98 + 149  = 446
id=3: 50 + 100 + 50 + 100       = 300
id=4: 48 + 98 + 47              = 193
```

The final ranking was therefore:

```
1st: id=2, 446 points
2nd: id=1, 443 points
3rd: id=3, 300 points
4th: id=4, 193 points
```

The checksum was: 1*446 + 2*443 + 3*300 + 4*193 = `3004`

```
For easier debugging of this rather complicated task,
one more example is available here.

The calculation file contains the ranking for every quest and every part,
as well as the final scores, ranking positions, and checksum.
```

Pay special attention to the checksum calculation. The final ranking may also contain ties. In the example, two participants finished in 55th place with 374 points, and the checksum includes: `... + 55*374 + 55*374 + 57*...` rather than `... + 55*374 + 56*374 + 57*...`.

What was the checksum of the final ranking?

Your notes for this part:
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Part III solved with answer: 7106835
Check your progress

Puzzle solved! Don't stop now!

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